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Studierichting: Informatica
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$$\text{Opgave: } z^6 = 1$$

$$r = \sqrt{1^2 + 0^2} = \sqrt{1} = 1$$

$$\tan \theta = \frac{0}{1} = 0 \Rightarrow \theta = \text{Bgtan } 0 = 0$$

$$\Rightarrow w = \sqrt[6]{1} \operatorname{cis} \left(\frac{0 + k \cdot 2 \cdot \pi}{6} \right)$$

$$w_0 = 1 \cdot \operatorname{cis}(0) = 1$$

$$w_1 = 1 \cdot \operatorname{cis} \left(\frac{2\pi}{6} \right) = \operatorname{cis} \left(\frac{\pi}{3} \right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$w_2 = 1 \cdot \operatorname{cis} \left(\frac{4\pi}{6} \right) = \operatorname{cis} \left(\frac{2\pi}{3} \right) = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$w_3 = 1 \cdot \operatorname{cis}(\pi) = -1$$

$$w_4 = 1 \cdot \operatorname{cis} \left(\frac{8\pi}{6} \right) = \operatorname{cis} \left(\frac{4\pi}{3} \right) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$w_5 = 1 \cdot \operatorname{cis} \left(\frac{10\pi}{6} \right) = \operatorname{cis} \left(\frac{5\pi}{3} \right) = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

References